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NiCoP-stabilized single atoms for efficient hydrogen evolution: A first-principles study

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ABSTRACT

The efficient production of hydrogen through water electrolysis can be regarded as a pragmatic strategy for the generation of clean, low-carbon, safe, and highly efficient energy. However, to overcome the significant overpotential, the sluggish hydrogen evolution reaction (HER) requires a significant energy input. A good atomic-level electrocatalyst design strategy plays a crucial role in improving HER performance. In this work, density functional theory (DFT) calculations are systematically applied to investigate single metal-atom electrocatalysts stabilized on NiCoP supports (M@NiCoP). Initially, a computational evaluation of the thermodynamic and electrochemical stability of all the catalysts was conducted. Subsequently, the hydrogen adsorption Gibbs free energies of these catalysts were calculated, to screen out materials that exhibited both good performance and stability. The research result indicates that Sn@NiCoP and Pt@NiCoP exhibited excellent stability and catalytic activity, especially the ΔG_{H^*} of Pt@NiCoP catalyst with only 0.01 eV (pH = 0), which outperforms the current Pt/C catalyst. This research offers theoretical predictions regarding the design of low-cost HER electrocatalysts. Moreover, it has pioneered new methods for stabilizing single-atom (SAC) catalysts.

1. Introduction

Since fossil fuels are non-renewable resources and a significant cause of environmental pollution, there is an urgent need to develop non-fossil fuel energy sources that are both renewable and clean [1]. Hydrogen possesses a high calorific value when combusted, its by-products are entirely devoid of pollutants, and there are substantial quantities of hydrogen-containing resources on the planet [2]. Consequently, it is regarded as an ideal clean energy. Among the various hydrogen production methods, water electrolysis has emerged as a particularly encouraging and appealing approach [3]. Efficient electrocatalysts are crucial in reducing the overpotentials generated by the retarded transfer of multiple proton-coupled electrons during the water electrolysis reaction [4]. At present, the predominant composition of HER electrocatalysts is Pt-based materials [5]. Nevertheless, several limitations, including limited resources, high expenses, and relatively short lifespans, impede the widespread application of these materials. Consequently, the design of highly efficient, cost-effective, and stable HER catalysts has become an urgent and pressing task.

In recent times, transition metal single-atom catalysts (TMSACs), with the unique feature of having only a single metal atom at their active sites, have found extensive application in water electrolysis [6,7]. Despite their ability to maximize atomic utilization and demonstrate exceptional efficiency in the hydrogen evolution reaction (HER), transition metal single-atom catalysts are prone to aggregation. This aggregation phenomenon results in a decline in catalytic activity [8]. The relatively high Gibbs free energy of individual atoms makes them prone to clustering into nanoparticles, further diminishing catalytic effectiveness. To address this issue, it is possible to leverage the strong interactions between single metal atoms and their supporting material to enhance catalytic activity. These robust interactions can confine the metal atoms. As a result, modulating the interaction between single metal atoms and their supports proves to be of utmost importance for optimizing the performance of catalysts. Noble metals and carbon-based

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substances are extensively employed as single-atom components and supporting materials, respectively, in the context of water electrolysis [9]. Liu et al. [10] found that the repair process can be used to anchor various transition metal atoms when heating polyhexaphenylbenzene materials with intrinsic pores, where loaded Ir SAC catalysts showed significantly higher electrocatalytic activity and stability in HER compared to commercial Pt/C and Ir/C. Thanh Duc Le et al. [11] have successfully synthesized a distinctive catalyst that concurrently anchors Pt and Ni diatoms within the nitrogen-doped carbon framework (NDCF). The high-surface-area, N-rich NDCF provides dense single-atom (Pt, Ni) active sites. This catalyst shows a 24.7 mV overpotential at 10 mA cm^{-2} and 6.1 A $mgPt^{-1}$ mass activity (at 40 mV), indicating its excellent HER electrocatalytic activity. Nevertheless, carbon-based materials characteristically exhibit instability under conditions involving elevated overpotentials or oxidative environments. This instability is the cause of the active metal detaching from the support material, which in turn results in catalyst deactivation [12]. As a consequence, the identification of carbonless and non-expensive metal single-atom catalysts is of paramount importance for the widespread industrial hydrogen production.

Transition metal phosphides (TMPs) have garnered substantial interest in electrocatalysis research, primarily attributed to their unique electronic configurations, metallic properties, and cost-efficient nature [13,14]. Compared to monometallic phosphides, bimetallic systems such as NiCoP [15,16] exhibit enhanced electrical conductivity and pronounced synergistic interactions between constituent elements, thereby optimizing charge transfer and catalytic performance. These properties culminate in enhanced electrocatalytic activity. Notwithstanding the significant advancements achieved, the performance of individual NiCoP remains less than ideal. This is mainly ascribed to the excessively strong interaction between cobalt (Co) and adsorbed hydrogen (H_{ads}). This interaction frequently impedes the desorption of H_{ads} to generate H_2. Therefore, their catalytic activity and operational stability fall short of industrial application standards, particularly in large-scale hydrogen production [17]. The latest research demonstrates that single-atom catalyst design holds significant promise in tailoring electronic structures, effectively reducing reaction barriers, accelerating reaction kinetics, and ultimately enhancing intrinsic activity [18,19]. Chen et al. [20] synthesized defect-rich NiCoP nanowire arrays via low-temperature phosphatization, followed by precise single-atom Ru anchoring through gradient annealing to form the high-performance catalyst Ru–NiCoP. DFT calculations confirmed that Ru-induced electronic structure modulation significantly reduced the dehydrogenation energy barrier, thereby enhancing catalytic performance.

Inspired by previous catalyst screening studies [21], this study innovatively devised a series of NiCoP-stabilized metal single-atom catalysts (M@NiCoP, where M = transition metal atoms), which represent a novel approach in the field, opening up new possibilities for enhancing catalytic performance and efficiency. For the first time, the stability and HER performance of these catalysts were systematically explored through density functional theory (DFT) calculations. Furthermore, volcano-type diagrams of the HER descriptors were constructed to select catalysts that exhibit superior performance in water electrolysis.

2. Computational details

The DFT calculations were carried out in the Vienna ab initio simulation package (VASP) [22] and the obtained results were subsequently post-processed with the Vaspkit package [23]. During the structure optimization and self-consistent calculations, the generalized gradient approximation (GGA) with the Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional [24], was employed to describe electronic interactions. For all plane-wave basis sets, a cutoff energy of 450 eV was set. The Brillouin zone was sampled using a 3 × 3 × 1 Monkhorst-Pack grid for the slab model. Structure optimization was carried out via the conjugate gradient (CG) algorithm, with Van der Waals interactions corrected by the DFT-D3 method [25]. Convergence criteria for energy and force were set at 1×10^{-6} eV and 0.02 eV $Å^{-1}$, respectively. Ab initio molecular dynamics (AIMD) simulations were performed using the canonical (NVT) ensemble with the temperature (298K) regulated by the Andersen thermostat [26]. The climbing image NEB (CI-NEB) method [27] was used to calculate transition states and determine the minimum energy paths (MEP) by optimizing a chain of images between the initial state and the final state. To comprehensively examine the impact of solvation effects, this study utilized the VASPsol program to establish an implicit solvent model for simulating the aqueous environment. The comparison of ΔG_{H^*} data with and without solvation effects considered has been updated in Table S3.

As shown in Fig. 1, we choose the low-index surface 001 of NiCoP as the substrate to support metal atoms and thus form M@NiCoP. To account for the effect of magnetism, ISPIN = 2 has been used for all calculations in this paper. To avoid interactions between adjacent grids, we constructed a vacuum layer with a height of 15 Å.

The hydrogen evolution reaction is a comparatively simple two-electron transfer process. It occurs via various reaction mechanisms, such as the Tafel and Heyrovsky mechanisms:

Volmer:

$$M + H_2O + e^- \rightarrow M - H^* + OH^- \quad (1)$$

Heyrovsky:

$$M - H^* + H_2O + e^- \rightarrow M + H_2 + OH^- \quad (2)$$

Tafel:

$$2M - H^* \rightarrow 2M + H_2 \quad (3)$$

Where M represents the surface of the water electrolysis catalyst, and M-H* denotes the hydrogen atoms adsorbed on the catalyst surface. Regardless of the pathway selected by the HER, the H* intermediate is a constant constituent of the HER process. Therefore, the Gibbs free energy of hydrogen adsorption will serve as an important parameter for assessing the electrocatalytic activity [28]:

$$\Delta E_H = \frac{1}{n}\left(E_{(surf+nH)} - E_{surf} - \frac{n}{2}E(H_2)\right) \quad (4)$$

$$\Delta G_{H_*} = \Delta E_H + \Delta E_{zpe} - T\Delta S_H \quad (5)$$

In this context, ΔE represents the hydrogen adsorption energy, $E_{surf + nH}$ is the total energy of n hydrogen atoms adsorbed on the catalyst surface, E_{H2} is the energy of gaseous hydrogen molecules, Esurf indicates the energy of the catalyst surface, ΔG_{H^*} is the free energy of hydrogen adsorption, ΔE_{zpe} is the zero-point energy correction, and $T\Delta S_H$ is the entropy correction. According to Sabatier's principle, as an ideal HER catalyst, the interaction force between the catalyst and the H* intermediates should be precisely balanced, neither too weak nor too strong. If the force is too weak, hydrogen adsorption onto the catalyst surface becomes difficult, impeding the formation of H* intermediates. Conversely, if the force is too strong, hydrogen desorption from the catalyst surface is hindered. This results in a large number of surface-active sites being occupied, leading to catalyst poisoning. An appropriately balanced force not only enhances charge and mass transfer but also enables the rapid desorption of H_2. Collectively, this can remarkably enhance the overall reaction rate of the HER.

3. Results and discussion

3.1. The structure and stability of the M@NiCoP

We use NiCoP as a support material to stabilize single metal atoms. In the initial stage of our research, an in-depth exploration of the structural stability of M@NiCoP was carried out. In the course of the examination of atom adsorption, all prospective sites (illustrated in Fig. S1) were

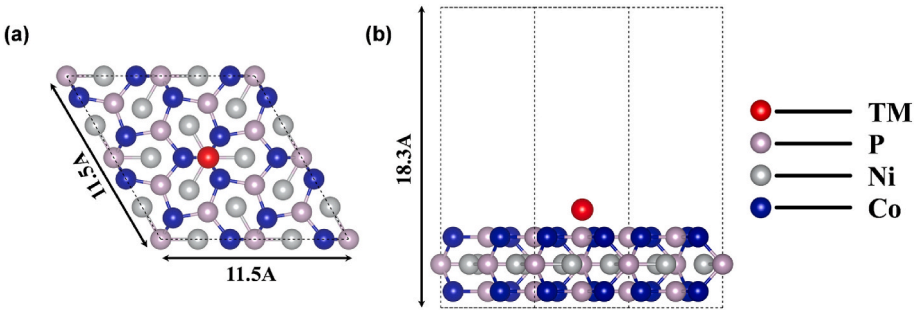


Fig. 1. The theoretical calculation models of M@NiCoP

incorporated into the analysis. Our findings reveal that the initial configurations (a) and (b) at the phosphorus (P) top position and cobalt (Co) top position respectively evolve and optimize into the nickel (Ni) top position (d) and triple-coordinated Co vacancy (c) states. As a catalyst for the hydrogen evolution reaction (HER), NiCoP encounters a crucial bottleneck due to the excessively strong adsorption of hydrogen intermediates (H*) at the cobalt active sites. This characteristic significantly hinders the reaction kinetics [29,30]. Theoretical investigations demonstrate that superatom position, characterized by unique coordination environments [31], effectively suppress metal atom migration and aggregation, thereby endowing materials with excellent structural stability. Building on this, this study systematically investigates the loading of 24 transition metal atoms onto superatom position (i.e., triple-coordinated Co vacancies) in NiCoP.

For an excellent catalyst, good stability is a fundamental requirement. In single-atom catalyst systems, it is important to avoid aggregation of individual atoms to prevent destabilization of the catalyst. The binding energy (E_b) and the cohesive energy (E_c) between the single atoms and the NiCoP support material were calculated in order to assess the structural stability of M@NiCoP. The comparison of the values of E_b and E_c is a common approach to determine whether SAC can effectively inhibit aggregation and dispersal. E_b and E_c are calculated as follows [32]:

$$E_b = E_{NiCoP} + E_{M-single} - E_{M@NiCoP} \tag{6}$$

$$E_c = E_{M-single} - E_{M-bulk}/N \tag{7}$$

Where $E_{M@NiCoP}$ represents the energy of M@NiCoP, $E_{M\text{-}single}$ represents the energy of a metal atom and E_{NiCoP} is the energy of NiCoP supporting material. $E_{M\text{-}bulk}$ and N represent bulk crystal unit cell energy and the number of atoms in a unit cell. If $E_b > E_c$, thermodynamically, the metal single atom has a preference for binding with the NiCoP supporting material rather than aggregating to form nanoclusters [33]. Based on previous work [34], we compared the values of E_c and $E_{c\text{-}exp}$ and found that the difference is small and the error is within reasonable limits. The E_c, $E_{c\text{-}exp}$, and $E_b - E_c$ values are listed in Table S1. Subsequently, the thermodynamic and electrochemical stabilities of all catalysts were assessed through the formation energy (E_f) and dissolution potential (U_{diss}). The calculation methods for these parameters are presented as follows [34]:

$$E_f = E_{M@NiCoP} - E_{NiCoP} - E_{M-bulk} \tag{8}$$

$$U_{diss} = U^0_{diss} - E_f/N^0 e \tag{9}$$

Where U^0_{diss} and N^0 denote the standard dissolution potential of bulk metal and the number of transferred electrons during the dissolution process, respectively. The U^0_{diss}, N^0, E_f and U_{diss} values are tabulated in Table S2. Based on the previous research [34], catalysts are regarded as thermodynamically stable when $E_f < 0$ eV. Meanwhile, a catalyst with $U_{diss} > 0$ V vs. the standard hydrogen electrode (SHE) is considered electrochemically stable.

As depicted in Fig. 2a – b, M@NiCoP (M = Au, Pt, Pd, Sn, Cd, Zn, Nb, In, Zr, Y, Ti, SC) has been found to exhibit favorable structural and thermodynamic stability characteristics. However, it is important to note that of the various catalysts considered, only the Au@NiCoP, Pt@NiCoP, Pd@NiCoP, Sn@NiCoP, and Cd@NiCoP catalysts were able to satisfy the structural, thermodynamic, and electrochemical stability standards simultaneously. This indicates that they can stabilize the composition of M@NiCoP and maintain excellent electrochemical stability during water electrolysis.

In order to further verify the thermodynamic stability of the aforementioned five catalysts, we carried out AIMD simulations on their 3 × 3 × 1 supercells. As depicted in Fig. S2, neither structural distortion, bond rupture, nor phase transition was detected. Nevertheless, upon observing the energy variation, it was noticed that there is a marginal increment in the energy of Cd@NiCoP, suggesting that it might exhibit insufficient stability at 298K. In addition, the total energy of the remaining four materials progressively stabilizes during the simulation process, with a relatively narrow fluctuation range. This indicates that these materials possess enhanced energy stability at this temperature, which consequently reflects their favorable thermodynamic stability. Based on previous studies [31], which suggest that metal single atoms on the NiCoP surface may migrate and form aggregates, this study systematically investigates the migration behavior of Pt atoms in the Pt@NiCoP system. To elucidate its dynamic stability, we employed the Climbing Image Nudged Elastic Band (CI-NEB) method to construct the migration pathway, with the Pt atom initially positioned at a surface superatomic site and the neighboring Ni top site A as the final configuration. The calculation results reveal that the transition state of the migration pathway is located at the Co–Co bridge site B, with a migration energy barrier of 0.98 eV (Figs. S3a–b). This significant energy barrier indicates that Pt atoms are unlikely to undergo long-range migration on the NiCoP surface, effectively suppressing the formation of aggregates. To further validate the kinetic stability of the material system, we performed ab initio molecular dynamics (AIMD) simulations at 298 K. Analysis of the 10 ps trajectory (Fig. S3c) shows that the Pt atom only exhibits minor vibrations around its equilibrium position, with no significant structural changes observed. Combining transition state theory and molecular dynamics simulations, these results confirm that the Pt@NiCoP system possesses excellent anti-migration stability at the atomic scale.

To comprehensively understand the surface properties of M@NiCoP, Bader charge analysis was employed to calculate the transferred charge upon the adsorption of a single metal atom on the NiCoP support.

As depicted in Fig. 2c and d, Platinum-group metals (including Pt, Au, Pd, Ir, and Rh atoms) carry positive charges. This indicates the transfer of electrons from these metal atoms to the NiCoP catalysts, with Pt demonstrating the most positive Bader charge of 0.23e. Furthermore, we calculated the charge difference density diagrams of Pt@NiCoP, as presented in Fig. 2f. The cyan region represents electron loss, while the yellow region indicates electron gain. It is observed that electrons are transferred from the NiCoP substrate to the Pt atoms, a result that

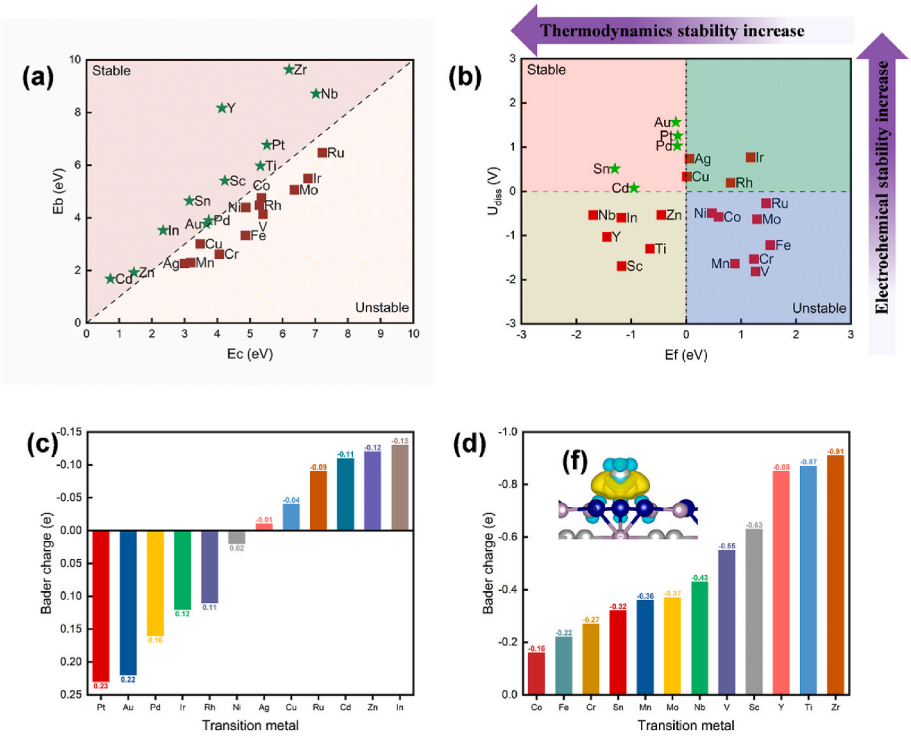


Fig. 2. (a) E_b and E_c of the metal atom in M@NiCoP. (b) E_f and U_{diss} of M@NiCoP. (c–d) Bader charge of metal single atoms adsorption on M@NiCoP.

corroborates with the Bader charge. Conversely, Cu and some other atoms bear negative charges. This is because electrons are transferred from the NiCoP-supporting material to the transition metal atoms. As a result, ionic bonds govern the interaction between single metal atoms and the NiCoP-supporting material. Furthermore, the transition metal atoms in Group IIIB to VB have relatively negative Bader charges, especially Zr@NiCoP has the most negative Bader charge of −0.91e. In particular, the Bader charges of Ni@NiCoP (0.02e) and Ag@NiCoP (0.01e) are very close to zero.

3.2. HER performance

Having confirmed the stability, the catalytic performance of M@NiCoP was investigated. The Gibbs free energy of hydrogen adsorption (ΔG_{H^*}) is a crucial factor that significantly impacts the HER performance. In general, when the value of ΔG_{H^*} is near zero, electrocatalysts have a tendency to display outstanding performance in the hydrogen evolution reaction [35]. A classic three-state plot can be employed to depict the process of HER. The ΔG_{H^*} values of M@NiCoP catalysts were calculated to conduct a qualitative evaluation of their

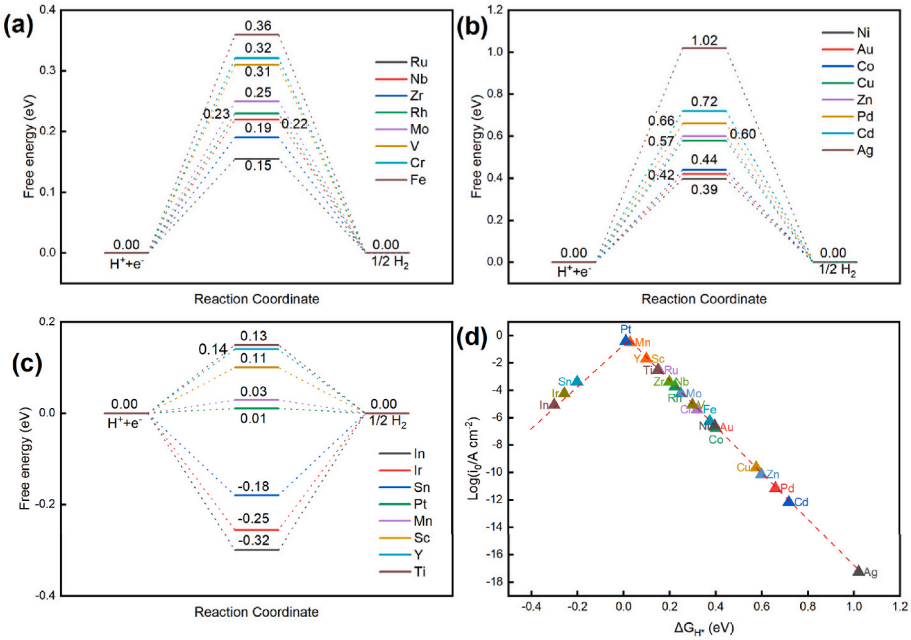


Fig. 3. (a–c) Free energy diagrams of HER. (d) the HER volcano curve relationship between the ΔG_{H^*} of the M@NiCoP and theoretical exchange current i_0.

HER performance.

As presented in Fig. 3a–c, the Ag@NiCoP catalyst exhibits the highest ΔG_{H^*} value of 0.99 eV. This indicates that the adsorption of hydrogen (H*) on Ag@NiCoP is not a kinetically favorable process. Consequently, the Ag@NiCoP catalyst demonstrates poor performance in the hydrogen evolution reaction. Electrocatalysts with $|\Delta G_{H^*}| < 0.20$ eV are regarded as promising candidates for the HER. The ΔG_{H^*} values of Y@NiCoP, Sc@NiCoP, Mn@NiCoP, Sn@NiCoP and Pt@NiCoP catalysts are 0.14eV, 0.13eV, 0.03eV, −0.18 and 0.01eV, respectively. In contrast to the ΔG_{H^*} of CoNiP, which is −0.63 eV [36], they have even more excellent catalytic activity. However, according to the binding energy calculations, Y@NiCoP, Sc@NiCoP and Mn@NiCoP have poor thermal stability. Therefore, these catalysts are not considered promising candidates for the hydrogen evolution reaction. A lower absolute value of $|\Delta G_{H^*}|$ promotes the release of hydrogen. This characteristic is advantageous for enhancing the HER activity. Sn@NiCoP and Pt@NiCoP could be excellent electrocatalysts for the HER because the ΔG_{H^*} value of Sn@NiCoP and Pt@NiCoP are close to zero, moreover, they are close to or lower than Pt (111) ($\Delta G_{H^*} = 0.09$ eV) [35].

A typical volcano curve emerges due to either the overly strong or weak adsorption of the intermediate. This phenomenon is not favorable to the performance of the HER. The theoretical exchange current (i_0) calculated according to Norskov's hypothesis was plotted on the HER volcano curve [37]. The exchange current at pH = 0 is expressed as:

$$i_0 = -ek_0 \frac{1}{1 + exp(|\Delta G_{H^*}/k_B T|)} \tag{10}$$

Where k_0 represents the rate constant, which was set to 1 under zero overpotential [38]. k_B and T are the Boltzmann constant and temperature, respectively. As depicted in Fig. 3d, the volcano diagram indicates that the M@NiCoP catalysts demonstrate superior HER catalytic performance when their exchange currents are close to the peaks of the volcano curve. When $\Delta G_{H^*}>0$, the adsorption of H* on the surface is comparatively weak. In this case, the i_0 increases dramatically as ΔG_{H^*} decreases. Conversely, when $\Delta G_{H^*}<0$, the adsorption of H* on the surface is relatively strong, and the i_0 decreases exponentially as ΔG_{H^*} decreases. Notably, the Pt@NiCoP catalyst is closest to the peak. As a result, it showcases the highest exchange current and outstanding HER catalytic activity. In addition, Sn@NiCoP, Mn@NiCoP, Sc@NiCoP, Ru@NiCoP, and Y@NiCoP are also in the vicinity of the peak, leading to their high exchange currents. This consequence is in good correspondence with the trend of ΔG_{H^*}. In Table 1, we systematically compared and analyzed the research progress of our study with existing literature on SACs for HER. These studies primarily focused on theoretical calculations of SACs, covering both theoretical predictions and experimental validation, with some theoretical predictions validated through experimental synthesis.

Table 1
Comparison of the electrocatalytic HER over single atom catalysts.

Catalyst	Overpotential (V_{RHE})	Ref.
Pt@NiCoP	0.01	This work
Sn@NiCoP	0.18	This work
Pt@VNC	0.05	[39]
IrP@NC	0.21	[40]
K_2PtCl_4@NC-M	0.04	[41]
V_2N_8	0.10	[42]
Pt-GT-FeCo/Cu	0.04	[43]
Ru@RuNx@GN/GN	−0.01	[44]
Ru@MHC	0.07	[45]
Ru/Na^+, K-PC	−0.06	[46]
CuRu@GN	0.08	[47]
IrCo	−0.07	[48]
RhCu/G_N	0.01	[49]
RhOOH/Rh	−0.24	[50]
Pt Cs/MoO_2	0.06	[51]
RuP_2@NPC	0.24	[52]

The Volmer-Tafel and Volmer-Heyrovsky reactions represent two possible routes for HER. It is commonly accepted that, during the Volmer process, protons in an acidic solution are rapidly adsorbed to form adsorbed H*. This step has been the subject of extensive research in the pursuit of a high-quality electrocatalyst. In order to gain a comprehensive understanding of the HER mechanism of the Pt@NiCoP and Sn@NiCoP catalysts, the subsequent stage, namely the generation of H_2, was further investigated as shown in Fig. 4. The results show that the kinetic barriers of the Tafel reaction for Sn@NiCoP and Pt@NiCoP are 1.16 eV and 0.43 eV, respectively, both of which are lower than the kinetic barriers of the Heyrovsky reaction, which are 2.01 eV and 1.08 eV. Thus, the Sn@NiCoP and Pt@NiCoP catalysts favor the Tafel reaction over the Heyrovsky reaction. Moreover, through comparison, it is evident that Pt@NiCoP exhibits the lowest Tafel energy barrier. This indicates that, in comparison to Sn@NiCoP, H^+ is more prone to combine with H* on the surface of Pt@NiCoP to produce hydrogen and be released. In conclusion, the Volmer-Tafel process is the primary pathway for the HER process in Sn@NiCoP and Pt@NiCoP catalysts due to the low energy barrier.

3.3. Electronic properties

Electrical conductivity is a critical factor in the efficiency of electron transfer during hydrogen evolution reactions. We calculated the density of states and the results are shown in Fig. S4. All M@NiCoP materials exhibit no bandgap, indicating decent electrical conductivity. Furthermore, the d-band center(ε_d) has been extensively utilized to comprehend and forecast the catalytic activity of metal surfaces. [53]. From the projected density of states (PDOS) of the d orbitals, the ε_d of the different active sites on the M@NiCoP catalyst can be deduced (Fig. 5a–d) [54]. The ε_d is given by Ref. [55]:

$$\varepsilon d = \frac{\int_{-\infty}^{\infty} p(\varepsilon)\varepsilon d\varepsilon}{\int_{-\infty}^{\infty} p(\varepsilon) d\varepsilon} \tag{11}$$

Here, $p(\varepsilon)$ represents the number of electrons in the corresponding d-band. It is a widely accepted proposition that the d-band centre is typically utilized to characterize the intensity of the interaction between adsorption intermediates and the catalyst surface. The interaction between the metal 3d orbital and the molecular orbital leads to the splitting of the molecular orbital energy levels, thereby forming low-bonding and high-antibonding orbitals [56,57]. It has been established that when the valence electron orbital of a metal atom is situated at a relatively elevated energy level, this results in the antibonding orbital of hydrogen atoms being elevated. In circumstances where the antibonding orbital exhibits a higher energy level than the Fermi level, a consequence of electron reflux, the antibonding orbital remains unoccupied. This situation is conducive to the adsorption process. Moreover, as the ε_d increases, the associated ΔG_{H^*} will also decrease.

As demonstrated in Fig. 5a, the Ag@NiCoP exhibited a low ε_d of −3.50 eV and demonstrated a weak interaction with hydrogen atoms, which was not conducive to hydrogen atom adsorption. Nevertheless, the introduction of Pt ($\varepsilon_d = -1.10$ eV), Mo ($\varepsilon_d = -0.41$ eV), and Mn ($\varepsilon_d = -0.61$ eV) resulted in the effective adjustment of the electronic configuration and the enhancement of the H adsorption by increasing the ε_d of the active sites in NiCoP. Consequently, the ΔG_{H^*} values of Pt, Mo, and Mn-doped NiCoP were found to approximate 0, which is advantageous in view of the enhanced compatibility between the H adsorption and desorption dynamic process.

We further established a quantitative correlation between hydrogen adsorption free energy (ΔG_{H^*}) and d-band center (ε_d) in M@NiCoP materials through linear regression analysis (Fig. 6). By calculating ε_d values at different transition metal-doped sites, a pronounced negative correlation was identified between ΔG_{H^*} and ε_d of dopant atoms (Pearson's r = −0.79). Specifically, systematic reduction in ΔG_{H^*} values was observed when the d-band center of doped metals shifted above the

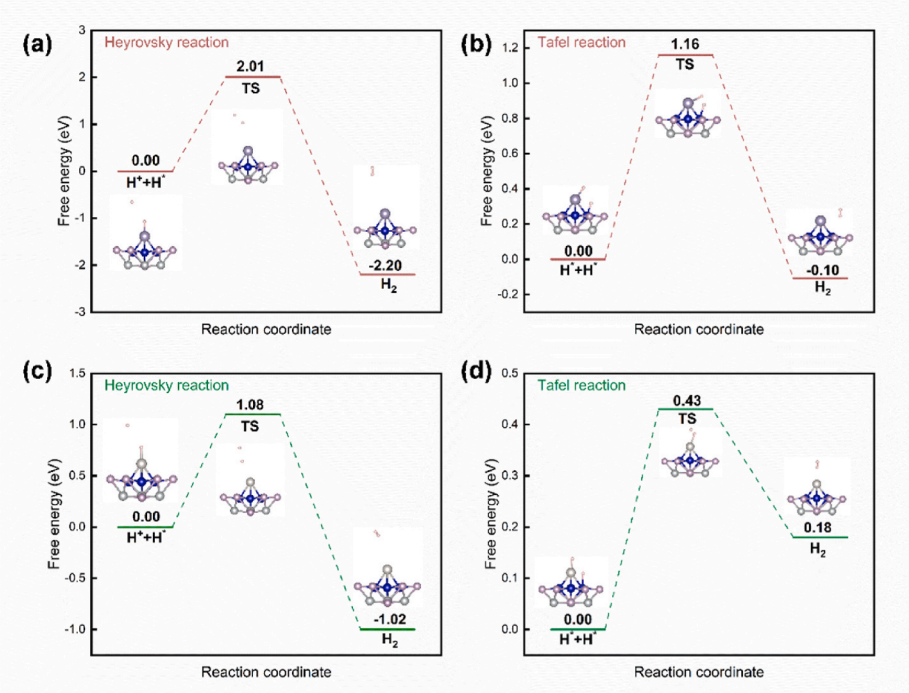


Fig. 4. Heyrovsky & Tafel reaction pathway for Sn@NiCoP(a-b), Pt@NiCoP(c-d)

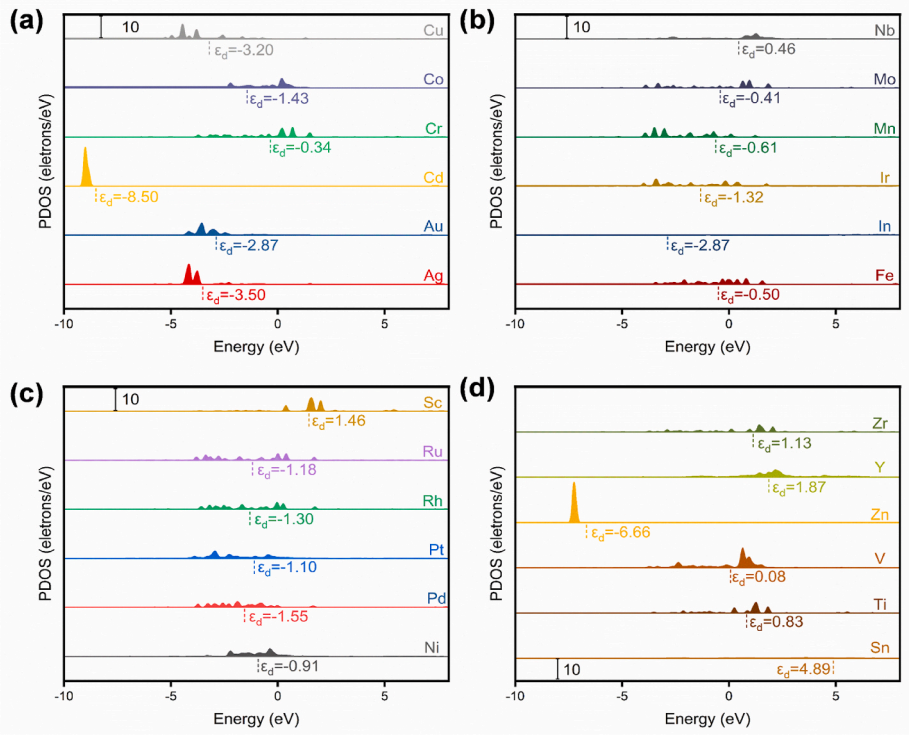


Fig. 5. (a–d) The projected density of states (PDOS) and ε_d of metal atoms adsorbed on the NiCoP.

Fermi level (elevated ε_d), indicating weakened adsorption strength of hydrogen intermediates. This linear relationship confirms that tailoring the electronic structure of transition metal dopants (e.g., selecting Ir with high ε_d or Zn with low ε_d) enables precise optimization of adsorption energetics for reactive intermediates at active sites, thereby significantly enhancing the hydrogen evolution reaction kinetics.

3.4. pH dependence of the HER

It is imperative to acknowledge the pivotal function of the pH value of the electrolyte solution in determining the ΔG_{H^*}. $\Delta G(pH)$ can be calculated as follows [58]:

$$\Delta G(pH) = k_b T \ln(10) \times pH \quad (12)$$

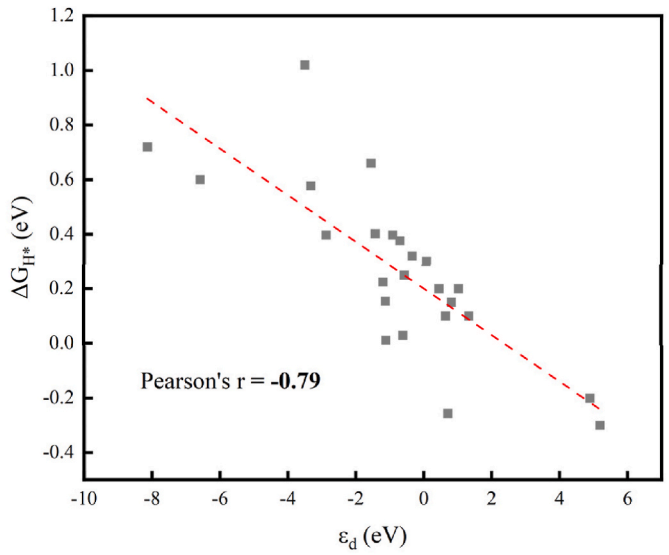


Fig. 6. Linear relationship between ΔG_{H^*} and ε_d of different mental sites.

As the pH value of the electrolyte solution increased, the value of ΔG_{H^*} increased. It is well-known that the closer the ΔG_{H^*} value is to 0, the more excellent the HER performance becomes. Therefore, it is essential to identify the appropriate pH range in which the M@NiCoP catalysts exhibit high catalytic activity. As demonstrated in Fig. 7, the ΔG_{H^*} values of the aforementioned models are displayed across a range of pH values from 1 to 14. As illustrated in Fig. 7a, only In@ NiCoP showed some HER performance between 2 and 8 pH, while the others exhibited poor HER performance. As shown in Fig. 7b, Mn@NiCoP and Pt@NiCoP exhibit high HER activity under strongly acidic conditions (pH = 0), with Pt@NiCoP demonstrating ΔG_{H^*} values comparable to commercial Pt/C. In contrast, Ir@NiCoP achieves optimal activity under weakly acidic conditions (pH = 4). This indicates that Mn@NiCoP and Pt@NiCoP systems are more suitable for strongly acidic environments, while the Ir@NiCoP displays pH-sensitive characteristics.From Fig. 7c, it can be seen that Sn@NiCoP exhibits higher HER activity in the pH interval of 2–6, while the ΔG_{H^*} of Sn@NiCoP is close to 0 when the pH is close to 3, which suggests that the catalyst prefers a acidic environment.

3.5. *Comparison with DFT + Machine learning approaches*

This study employs DFT calculations to provide atomic-scale insights into critical information such as electronic structures, adsorption energies, and reaction pathways, which are essential for deepening our understanding of the underlying mechanisms of single-atom catalysts. Notably, recent research on DFT- Machine Learning hybrid frameworks has demonstrated that machine learning techniques, through their predictive capabilities for key descriptors like ΔG_{H^*} and ε_d, can efficiently screen potential catalyst candidates across broad chemical spaces while significantly reducing computational costs [59,60]. These methods typically utilize machine learning models trained on DFT datasets to pre-screen materials, enabling rapid identification of high-performance candidates before targeted DFT validation. In contrast, this research focuses specifically on 24 types of M@NiCoP. While this focused strategy ensures rigorous validation of material stability and activity, it also restricts the scalability of screening. It is important to emphasize that the performance of machine learning models is highly dependent on the quality and diversity of training data, and their interpretability still lags significantly behind direct DFT-based mechanistic analysis. Nevertheless, future studies can further integrate density functional theory and machine learning methods to balance computational efficiency and analytical precision, particularly in the field of large-scale catalyst design, where this integration holds significant promise.

4. Conclusions

This study constitutes a comprehensive theoretical investigation of transition metal doped NiCoP for the electrocatalytic HER. This investigation was undertaken via DFT calculations. A systematic investigation was conducted into the crystal structure stability and the electronic properties, in addition to the ΔG_{H^*} of M@NiCoP. The findings suggest that the ΔG_{H^*} values of Sn@NiCoP and Pt@NiCoP approach zero, classifying them as exceptional HER electrocatalysts. Among them, Pt@NiCoP exhibits more excellent hydrogen evolution reaction (HER) performance under strongly acidic conditions, while Sn@NiCoP is more suitable for weakly acidic environments. In addition, this work found that the ε_d of the active sites of these 24 kinds of M@NiCoP catalysts has a decent linear relationship with their ΔG_{H^*}, which implies that more efficient catalysts can be designed by adjusting their ε_d. This study demonstrates that the incorporation of dopants is an advantageous method to enhance the electrocatalytic efficiency of HER. Selecting suitable dopant atoms facilitates the design of high-performance hydrogen evolution catalysts for practical applications.

CRediT authorship contribution statement

Yihong Sun: Writing – original draft, Software, Formal analysis, Data curation, Conceptualization. **Jianfeng Wan:** Supervision, Conceptualization. **Wenyan Bi:** Supervision, Investigation. **Shizheng Xie:** Data curation. **Menglin Yu:** Software, Data curation. **Yikai Hou:** Visualization, Validation. **Tianen Li:** Investigation. **Dongkui Zhou:** Software. **Lebin Li:** Writing – review & editing. **Baozhong Liu:** Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

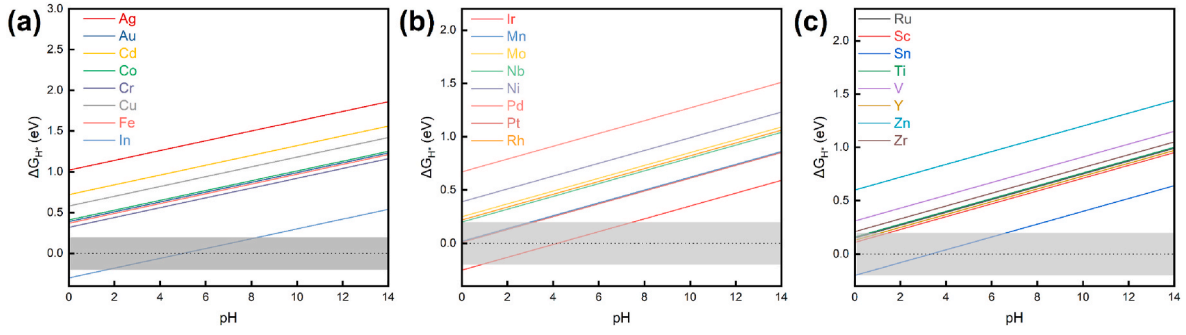


Fig. 7. The pH-dependent ΔG_{H^*} on different M@NiCoP. The range of −0.2 to 0.2 eV is considered to represent the optimal performance of HER.

the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijhydene.2025.03.323.

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